

KAIRUI BAO

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EDUCATION

University of Oxford

Master of Science (Major: Statistics)

Oxford, United Kingdom
Sep 2025 - Sep 2026(Expected)

University of Toronto (St. George Campus)

Honours Bachelor of Science with High Distinction (Major: Mathematics & Statistics)

Toronto, Canada
Sep 2021 - June 2025

CGPA: 3.92/4.0, Ranking: Top 1%

Awards: Dean's List Scholar 2023 Winter, Dean's List Scholar 2024 winter, Dean's List Scholar 2024 summer
New College Council In-Course Scholarship \$1000 (Top1% in college)

PUBLICATION

Luai Al Labadi, Omid Aghababaei jazi, Masoud Ataei, **Kairui Bao**, Ruodie Yu, *AI Meets Academia: Enhancing Statistics Education with ChatGPT in Undergraduate Courses*, Springer: TechTrends

Luai Al Labadi, Ruodie Yu, **Kairui Bao**, *Goodness-of-Fit Tests via Entropy-Based Density Estimation Techniques*, MDPI: Stats (All authors have contributed equally on this work)

Luai Al Labadi, Ayman Alzaatreh, Ruodie Yu, **Kairui Bao** *Goodness-of-Fit Tests via Kernel Density Estimation: Univariate and Multivariate Approaches*, JMLR (Under Review)

RESEARCH EXPERIENCE

Goodness-of-Fit Tests via Kernel Density Estimation: Multivariate Approaches

June 2025 – Oct 2025

Research Assistant, Supervisor: Prof. Luai Al Labadi (University of Toronto)

- Developed new goodness-of-fit tests based on kernel density estimation for both univariate and multivariate settings.
- Applied boundary correction using the reflection method to mitigate edge bias; accounted for bandwidth matrix selection and geometric invariance in the multivariate setting. The theoretical results demonstrated that the proposed tests are consistent and asymptotically converge to the Kullback–Leibler divergence under fixed alternatives.
- Extensive simulations and real data applications confirm the strong empirical performance of the tests, particularly in detecting smooth deviations from the null distribution.

Advancements in Estimation of Entropy for Model Assessment.

Jan 2025 – June 2025

Research Assistant, Supervisor: Prof. Luai Al Labadi (University of Toronto)

- Developed an entropy estimation algorithm based on the m-spacing density estimator, and Correa's estimator. Using extensive data simulations to visualize its improvements over the traditional Lagrange multiplier method; confirmed that the spacing estimator generally achieves superior predictive performance.
- Assessed the finite-sample performance of the proposed test statistics through simulation studies and applications to real data (Yarn Strength and River Flow) to validate the robustness of the proposed model.
- Applied theoretical proofs to derive the likelihood ratio test statistic for the m-spacing density estimator and its independence properties.
- Conducting power analysis under several alternative distributions to further evaluate the effectiveness of the proposed tests in detecting model misspecification.

ChatGPT-Driven Education: A Student-centered Approach

Sep 2024 - Feb 2025

Research Assistant, Supervisor: Prof. Luai Al Labadi (University of Toronto)

- Conduct an in-depth literature review to gather existing research on the integration of artificial intelligence, specifically ChatGPT, in educational settings.

- Use Bonferroni, Sidak-Bonferroni and other experimental design methods to investigate on the ChatGPT application on students' learning performance with other traditional teaching techniques.
- Analyze quantitative data using ANOVA, Mann whitney test and T-tests, performing a comparative analysis to identify any significant learning outcomes between the control and experimental group.
- Integrate quantitative and qualitative data to get holistic understanding of ChatGPT's impact on learning outcomes and student engagement.

The Application of 3D Object Detection Using Point Cloud

May 2024 - Aug 2024

Research Assistant, Supervisor: Prof. Jun Xiao (Zhejiang University)

- Investigated the basic operation and advantages of point clouds for 3D object recognition.
- Trained for point cloud recognition through a database.
- Tried to apply the algorithm structure of point net to the intelligent driving system of automobile.

The Application of Artificial Intelligence in Biopharmaceuticals

Jun 2023 - Sep 2023

Research Assistant, Supervisor: Prof. Yu Tian (Zhejiang University)

- Read and organized a large amount of literature on biological mega-health, including biopharmaceuticals, clinical medicine, patient database integration, and robotic surgery.
- Conducted methodological analysis of the application of artificial intelligence in various fields such as drug development, case information collection and retrieval, clinical surgery, CT diagnosis, etc.
- Exchanged experience with hospital doctors on disease pre-diagnosis and data analysis through data from large hospital databases.

SELECTED PROJECTS

Alzheimer's Disease identification

Sep 2024 - Dec 2024

- Used shrinkage method of Lasso to perform variable selection, and applied cross-validation through 10-fold to check the estimated mean squared error.
- Made comparison between Random Forest, Support Vector Machine, XGBoost, CatBoost and Decision Tree to train the encrypted Alzheimer's Disease dataset.
- Obtained a highly accurate model for identifying Alzheimer's Disease by continuous iteration and optimization, and held a Kaggle competition to rank the accuracy of different students in training large models.

Statistic Modelling of House Pricing in Iowa State in United States

Sep 2023- Dec 2023

- Utilized a multiple linear regression model to forecast the sale prices of housing in Ames.
- Import the Ames dataset into R and cleaned the data by removing the null observations, selected 9 predictors and performed preliminary data exploration of these predictors, and plotted the scatterplot matrix to check whether there is any non-linear relationship between each pair of predictors, and used the T-test and partial F-test to check the assumptions.
- Created a test model by replicating the final model and used the testing data to check the assumptions of our test model, and determined if it was the best-fitted model by analyzing the generated graphs.

Influence of the Composition of a Novel on Reader's Satisfaction from Multiple Perspectives

Sep 2021 - Dec 2021

- Investigated what factors can lead to a commonly liked attractive detective story in modern days to find the taste of detective fans and provide some evidence for future writers who want to write detective novels.
- Applied three different models to investigate how sufficient clues, sufficient justifications, difference in number of text words before and after the reveal and the main focus of the novel will influence the satisfaction level.

INTERNSHIP

Zhejiang Kai Chuang Environment Technology Company

Hangzhou, China

Research Assistant of the Research and Development Department

May 2022 - Sep 2022

- Participated in water quality testing experiments, and constructed the math model for the water treatment plant.
- Predicted the time and dose of compounds and microorganisms required at each wastewater treatment step through mathematical modeling.
- Used the Lasso method of machine learning to design a model for sewage treatment, which helped the company to get the bidding project of sewage treatment.

SKILLS

- Programming Languages: Python, R, Java, JavaScript, SQL, Matlab
- Tools: PySpark, Pandas, NumPy, SciPy, Scikit-learn, Keras, TensorFlow, PyTorch